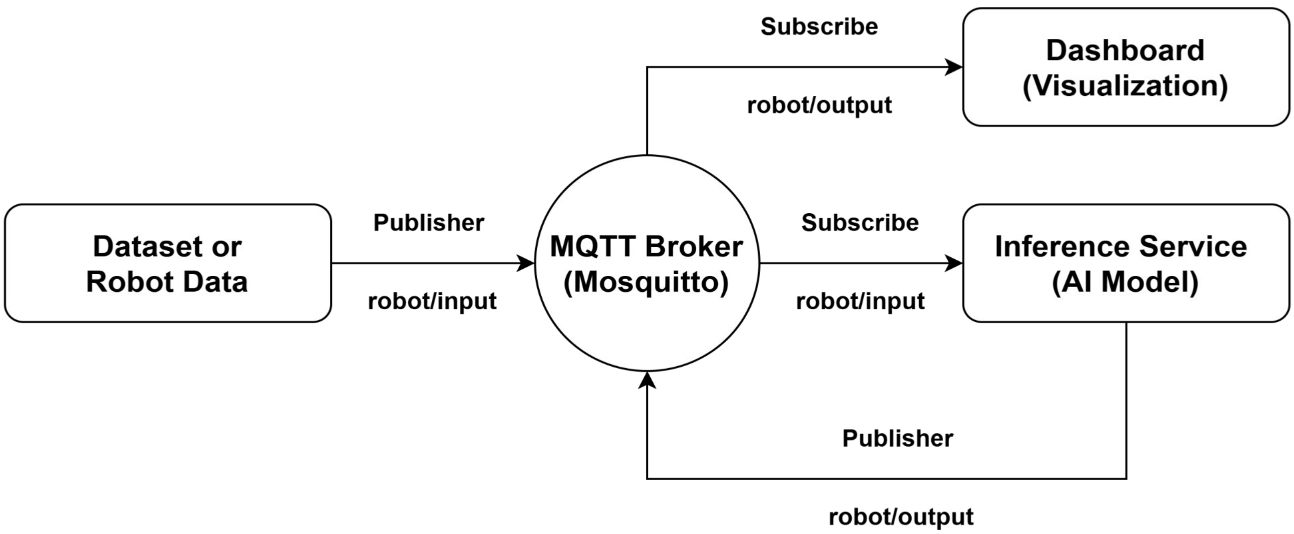
MQTT-Based AI Inference System for Real-Time Robotic Arm Health Monitoring

Objective

The objective of this module is to implement a real-time anomaly detection pipeline for robotic arm health monitoring as part of the Digital Twin predictive maintenance system.
The system receives robot telemetry data, preprocesses it, performs AI inference using a trained model, and provides real-time health assessment.

System Overview

The implemented architecture follows an event-driven pipeline using MQTT.



Communication Architecture

Protocol – MQTT Publish–Subscribe model
Topics

- Input Topic: robot/input
- Output Topic: robot/output

The publisher continuously streams robot telemetry to the inference service.
The inference service processes incoming messages and publishes diagnostic results.

Data Processing

Robot telemetry includes:

- Actual Current
- Target Current
- Joint Temperatures
- Target Moment

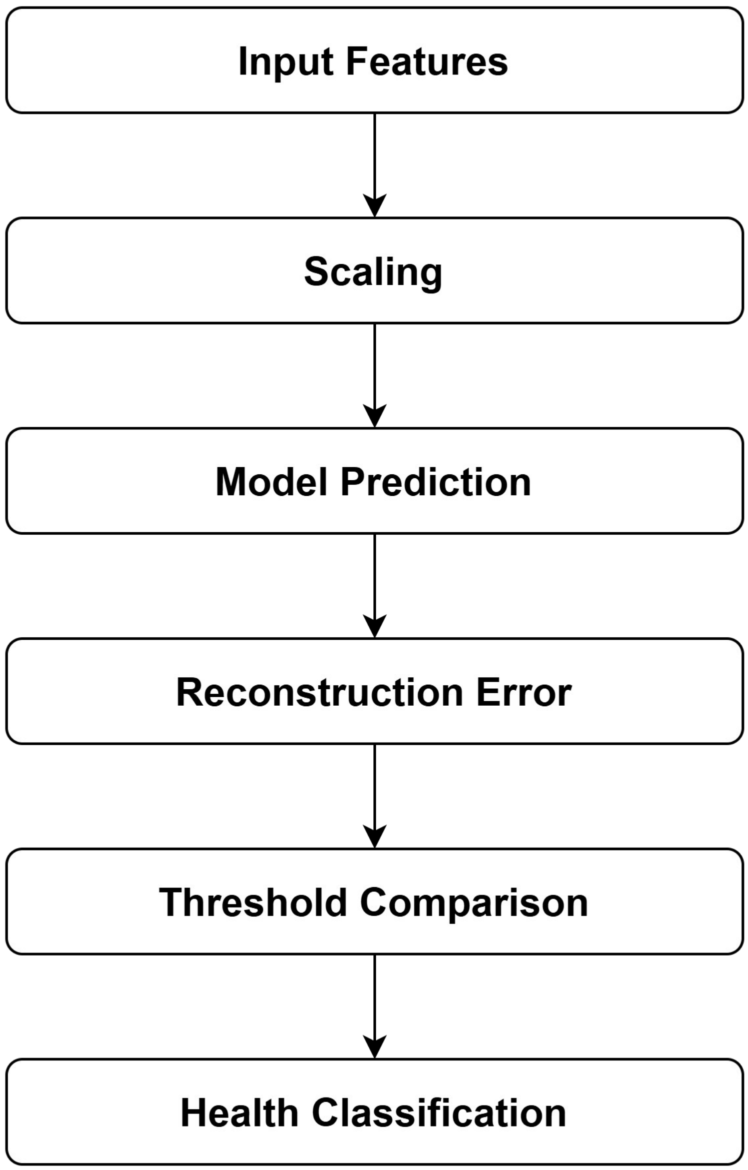
Derived feature: $\text{Track}_{Error} = \text{Actual}_{Current} - \text{Target}_{Current}$

Final model input:

- Actual Current
- Joint Temperatures
- Target Moment
- Current Tracking Error

AI Inference Module

Inference Flow:



Anomaly Score – Mean Squared Error (MSE)

Health States:

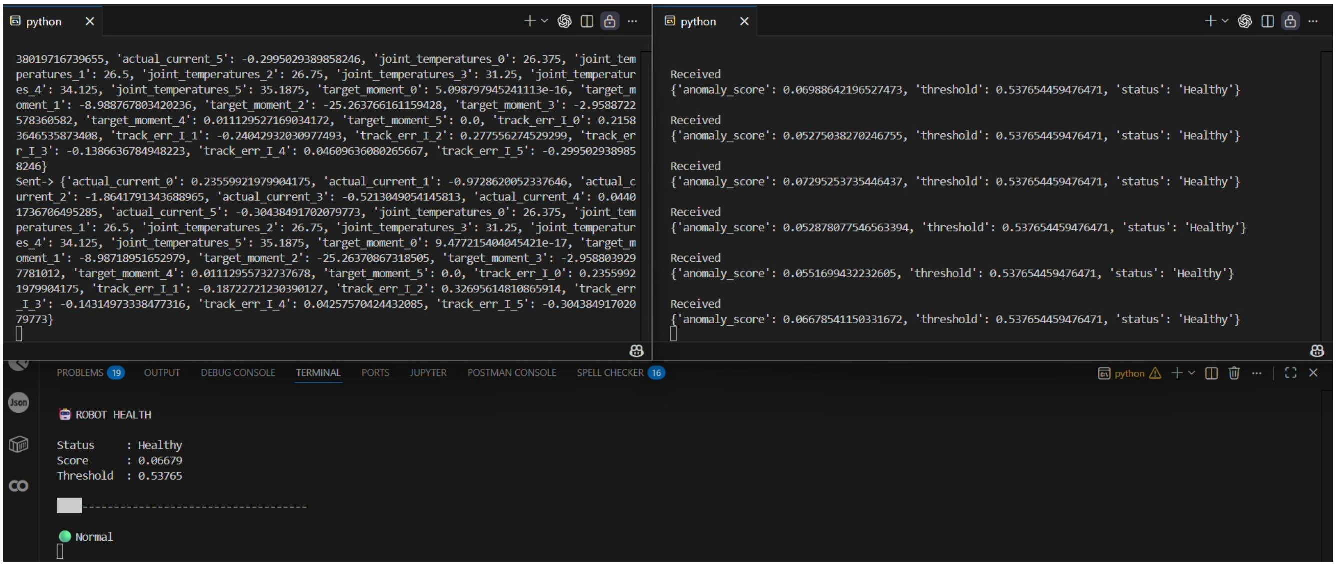
- Healthy → Score ≤ Threshold
- Critical → Score ≥ Threshold

Real-Time Monitoring Dashboard

A lightweight terminal dashboard was implemented. Displayed Information:

- Current Health Status
- Anomaly Score
- Threshold Value
- Health Indicator Bar

The dashboard updates continuously after receiving each prediction.



Module Description

File	Module Name	Responsibility	Input	Output
publisher.py	Telemetry Publisher	Simulates robot telemetry streaming by loading dataset records, performing feature extraction and publishing processed data to the MQTT input topic	CSV Dataset	MQTT Messages (robot/input)
inference_service.py	AI Inference Service	Receives telemetry data, preprocesses input features, performs AI inference using the trained model and generates health status predictions	MQTT Input Topic	MQTT Output Topic (robot/output)
dashboard.py	Health Monitoring Dashboard	Subscribes to prediction results and visualizes robot health condition in real time	MQTT Output Topic	Terminal Visualization

Future Work (Unity Integration)

As a future extension, the system will integrate a Unity-based digital twin to simulate robot behavior and generate live telemetry data. The generated data will be transmitted through MQTT to the Inference Service for processing and analysis, and the results will be displayed through health visualization components.